

EE351

微机原理与微系统

姜俊敏

电子与电子工程系

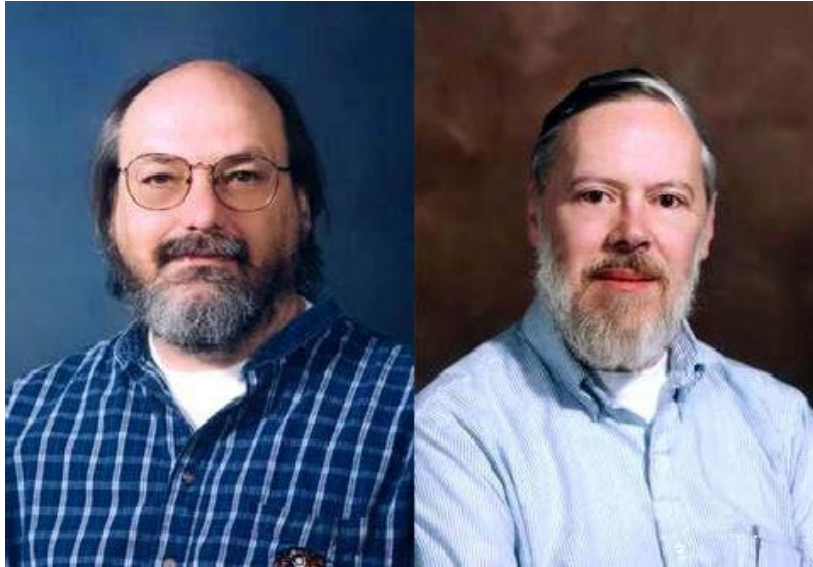
第三章 Introduction to Unix



Early History of Unix

- * 1960s
 - * Multics project (MIT, GE, AT&T)
- * 1970s
 - * AT&T Bell Labs
- * 1970s/80s
 - * UC Berkeley
- * 1980s
 - * DOS imitated many Unix ideas
 - * Commercial Unix fragmentation
 - * GNU (GNU's Not Unix) project

The Inventors



Ken Thompson Dennis Ritchie
ACM Turing Award winners, 1983

C Language as a Byproduct

- * Problem

- * Need to port Unix to other platforms
- * Unix written in assembly language

- * Solution

- * Rewrite Unix using 'B'
- * Extensive changes made for Unix

- * Renamed to 'C'

- * High & low level programming language
- * Increased portability
- * Easier to improve & enhance Unix

The Rebels



Richard Stallman who founded
Free Software Foundation



Linus Torvalds, father of
LINUX

A Post Seen in comp.os.minix

From: torvalds@klaava.Helsinki.FI (Linus Benedict Torvalds)

Newsgroups: comp.os.minix

Subject: Gcc-1.40 and a posix-question

Message-ID: <1991Jul3.100050.9886@klaava.Helsinki.FI>

Date: 3 Jul 91 10:00:50 GMT

Hello netlanders,

Due to a project I'm working on (in minix), I'm interested in the posix standard definition. Could somebody please point me to a (preferably) machine-readable format of the latest posix rules? Ftp-sites would be nice.

- * Two years after this post, there were 12,000 Linux users.

The chart illustrates the lineage of Unix-like operating systems, categorized by source type: Open Source (green), Mixed/Shared Source (orange), and Closed Source (pink). The timeline spans from 1969 to 2010.

Legend:

- Open Source (Green)
- Mixed/Shared Source (Orange)
- Closed Source (Pink)

Key Operating Systems and Lineage:

- Unics** (Mixed/Shared Source) leads to **Unix TSS 1 to 4**, **Unix TSS 5 to 6**, and **Unix TSS 7**.
- Unix TSS 1 to 4** and **Unix TSS 5 to 6** lead to **BSD 1.0 to 2.0** and **BSD 3.0 to 4.1**.
- Unix TSS 7** leads to **Unix 32v**.
- BSD 1.0 to 2.0** leads to **BSD 4.2** and **BSD 4.3**.
- BSD 3.0 to 4.1** leads to **BSD 4.2** and **BSD 4.3**.
- Unix 32v** leads to **Unix TSS 8** and **Unix TSS (Time Sharing System) 9 to 10**.
- Unix TSS 8** leads to **Unix TSS (Time Sharing System) 9 to 10** and **BSD 4.3 Tahoe**.
- Unix TSS (Time Sharing System) 9 to 10** leads to **BSD 4.3 Tahoe** and **BSD 4.3 Reno**.
- BSD 4.3 Tahoe** and **BSD 4.3 Reno** lead to **BSD 4.4 to 4.4 lite 2**.
- BSD 4.4 to 4.4 lite 2** leads to **Free BSD 1.0 to 2.2.x**, **Free BSD 3.0 to 3.2**, and **Free BSD 3.3 to 8.0**.
- Free BSD 1.0 to 2.2.x** leads to **Free BSD 3.0 to 3.2** and **Free BSD 3.3 to 8.0**.
- Free BSD 3.0 to 3.2** leads to **Free BSD 3.3 to 8.0**.
- Free BSD 3.3 to 8.0** leads to **OpenBSD 1.0 to 2.2** and **OpenBSD 2.3 to 4.x**.
- OpenBSD 1.0 to 2.2** leads to **OpenBSD 2.3 to 4.x**.
- OpenBSD 2.3 to 4.x** leads to **OpenBSD 5.0 to 5.0.4** and **OpenBSD 5.0.5 to 5.0.7**.
- OpenBSD 5.0 to 5.0.4** leads to **OpenBSD 5.0.5 to 5.0.7**.
- OpenBSD 5.0.5 to 5.0.7** leads to **OpenBSD 6.0**.
- OpenBSD 6.0** leads to **OpenBSD 7.x**.
- OpenBSD 7.x** leads to **OpenBSD 8.x**.
- OpenBSD 8.x** leads to **OpenBSD 9.x**.
- OpenBSD 9.x** leads to **OpenBSD 10.x**.
- OpenBSD 10.x** leads to **OpenBSD 11.x**.
- OpenBSD 11.x** leads to **OpenBSD 12.x**.
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- OpenBSD 91.x** leads to **OpenBSD 92.x**.
- OpenBSD 92.x** leads to **OpenBSD 93.x**.
- OpenBSD 93.x** leads to **OpenBSD 94.x**.
- OpenBSD 94.x** leads to **OpenBSD 95.x**.
- OpenBSD 95.x** leads to **OpenBSD 96.x**.
- OpenBSD 96.x** leads to **OpenBSD 97.x**.
- OpenBSD 97.x** leads to **OpenBSD 98.x**.
- OpenBSD 98.x** leads to **OpenBSD 99.x**.
- OpenBSD 99.x** leads to **OpenBSD 100.x**

Prevailed/Prevailing Systems

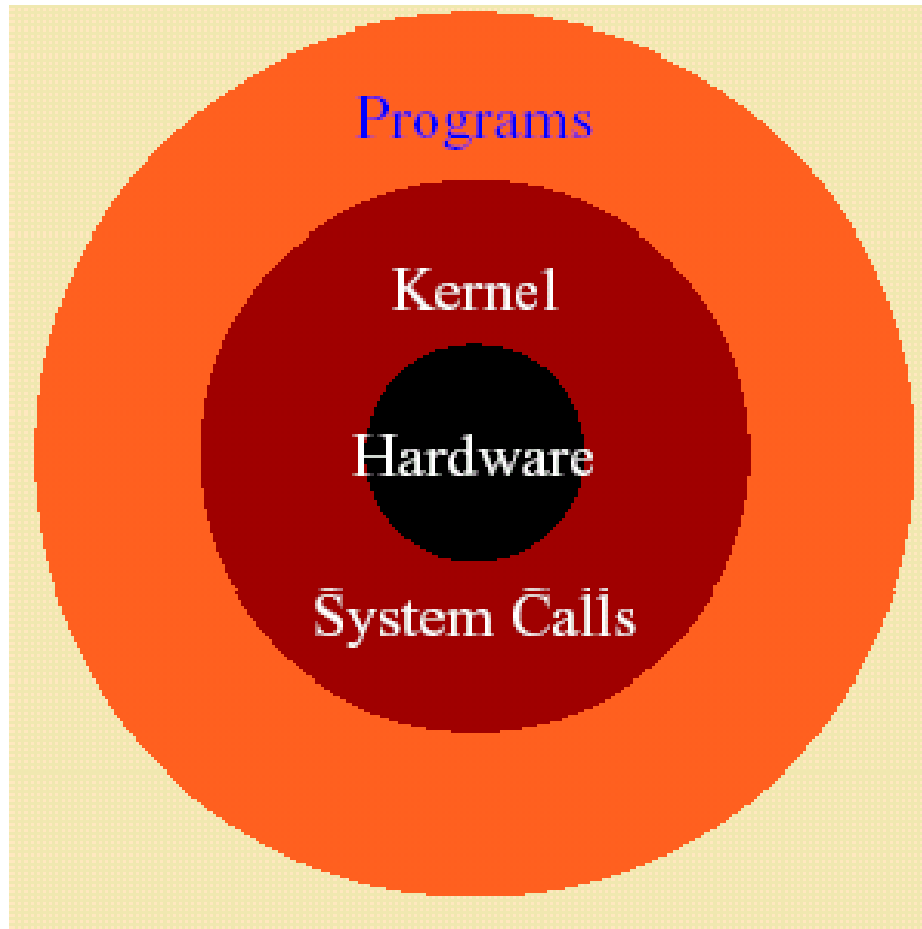
- * Unix is widespread and available from many sources, both free and commercial
- * [SunOS/Solaris/Open Solaris](#) - Sun Microsystems -> Oracle
- * [Digital Unix \(Tru64\)](#) - Digital -> Compaq -> HP
- * [HP-UX](#) - Hewlett Packard
- * [Irix](#) - SGI
- * [AIX](#) - IBM
- * [MacOS X](#) - Apple
- * [NetBSD/FreeBSD/OpenBSD](#) - UC Berkeley / the Net
- * [Linux](#) - Linus Torvalds / the Net
 - * Redhat/Fedora/CentOS
 - * Ubuntu
 - * Debian, SuSE, Mandrake..., etc.



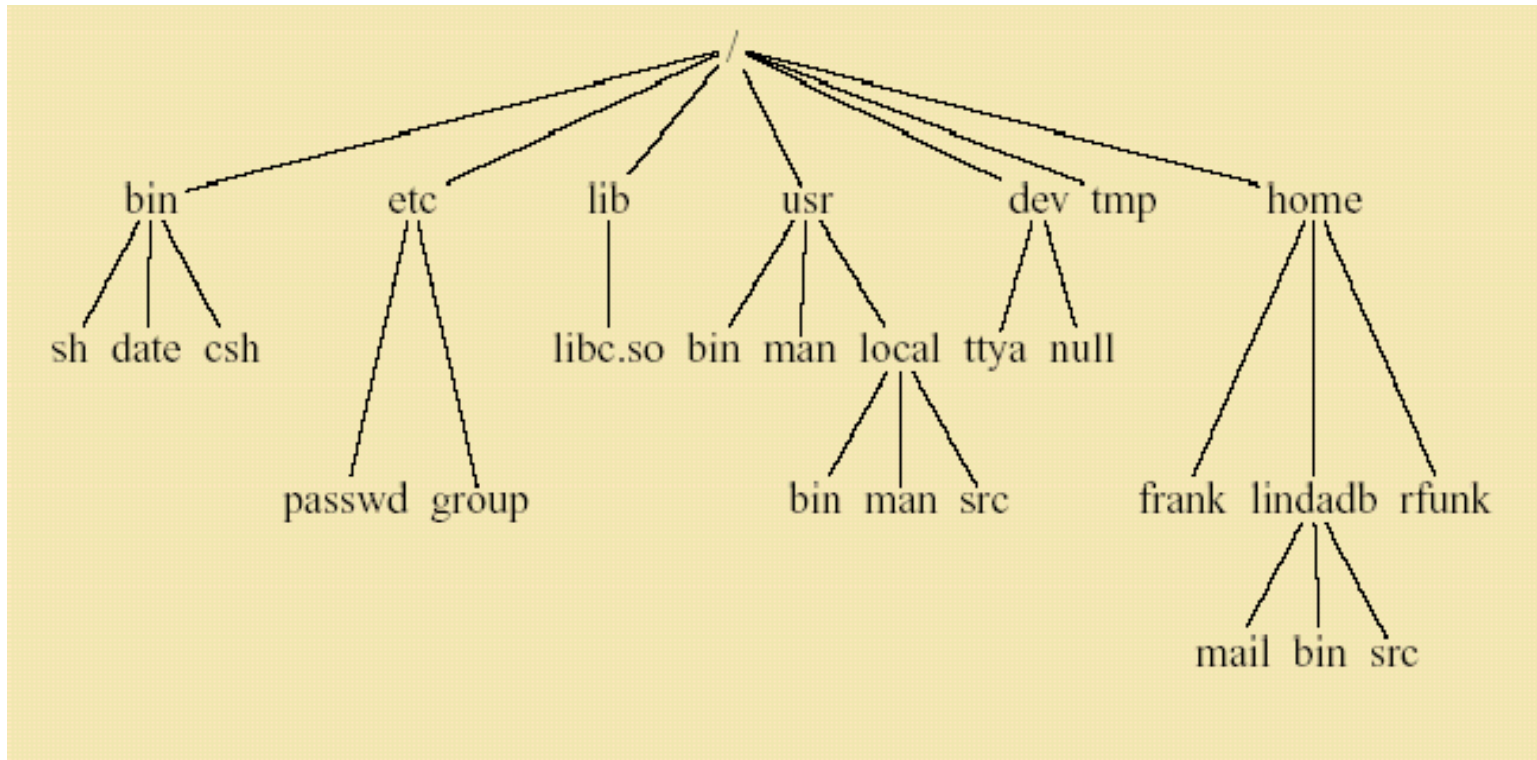
Unix Philosophy

- * **Multiuser / Multitasking**
- * Toolbox approach (make each do one thing well)
- * Flexibility / Freedom
- * Conciseness (small is beautiful)
- * **Everything is a file**
- * **File system has places, process has life**
- * Designed by programmers for programmers
 - * Even of the programmers

The Operating System Structure



The File System



Unix Programs

- * Shell is the command line interpreter
- * Shell is just another program
- * A program or command interacts with the kernel may be any of:
 - * built-in shell command
 - * interpreted script
 - * compiled object code file

Command Line Structure (1)

- * A command is a program that tells the Unix system to do something. It has the form:

command **options** **arguments**

- * 'Whitespace' separates parts of the command line
- * An *argument* indicates on what the command is to perform its action
- * An *option* modifies the command, usually starts with '-'

Command Line Structure (2)

- * Not all Unix commands will follow the same standards
- * Options and syntax for a command are listed in the 'man page' for the command
- * Be aware that Unix is case sensitive
 - * Use lower case
- * To terminate a command
 - * **^C** interrupt
 - * **^D** can log a user off; frequently disabled

Text-based and GUI-based

- * Historical reason
 - * tty/modem based terminal
- * Why text on command prompt mode?
 - * Efficiency for engineering
 - * Record keeping
 - * And etc.
- * Desktop use vs. server use
- * Experienced user vs. ordinary people

Getting Started



Login

- * Your user name
- * Your password
- * Control keys (^) perform special functions
 - * ^H or Backspace erase a character
 - * ^U cancel line
 - * ^S pause display
 - * ^Q restart display
 - * ^C cancel operation
 - * ^D signal end of file
 - * ^V treat following control character as normal character

Login/Logout Commands

- * Commands:
- * `login`
 - * `logout`
 - * `exit`
- * Snapshots of login GUI on Solaris and Ubuntu



Find Version of Unix

* `uname [options]`

```
[jiangjunmin@eex001 ~]$ uname -a  
Linux eex001 3.10.0-1160.25.1.el7.x86_64 #1 SMP Wed Apr 28 21:49:45 UTC 2021 x86_64 x86_64 x86_64 GNU/Linux
```

Change Password

- * **passwd**

- o l d passwd :

- new passwd :

- r e t y p e new passwd :

- * **Warning:** don't change your passwd unless the system administrator permits doing so

System Date and Time

- * `date`

Fri Jul 28 14:44:15 CST

- * System administrator can use 'date' to change the system time
- * '`man date`' for more details

Set User Environment (1)

- * **stty** - terminal control
 - * reports or sets terminal control options configures aspects of I/O control
 - * syntax: **stty** **attribute** **value**
 - * example: `stty erase ^H; stty kill ^U;`
- * Environment variables
 - * **HOME**
 - * **PATH**
 - * **SHELL**
- * **env** - show current EV values

Set User Environment (2)

- * **setenv** (csh)

- * `setenv PATH .:/bin:/bin:/usr/bin`
- * `setenv MANPATH /usr/share/man:/usr/local/man`
- * `setenv MANPATH $HOME/man:$MANPATH`

- * **set** and **export** (sh, bash)

- * `PATH=/usr/local/bin:/home/jack/bin; export PATH`
- * `export PATH=$PATH:/opt/acroread/bin`

- * **alias**

- * syntax: **alias** **aname** **pname**
- * `alias dir 'ls -al | more'`
- * `alias listc 'ls -al *.c | more'`

A .bashrc Example (1)

A terminal window with a blue title bar containing the text "jiangjunmin@eex001:~". The terminal background is dark gray, and the text is in a light blue/cyan monospace font. The content of the .bashrc file is displayed, showing shebang, global definitions, user-specific aliases, and license file sourcing.

```
#!/bin/bash

# Source global definitions
if [ -f /etc/bashrc ]; then
    . /etc/bashrc
fi

# User specific aliases and functions
alias fz='filezilla &'
export PATH=/app/local64/filezilla-3.5.3/bin:/app/local64/./app/local64/rar/:$PATH
alias df='df -h'
alias i='icfb &'
alias epdf="evince"
alias vcs='vcs -full64'
#
export APP_DIR='/app'

# License file
source /app/config/LICS/bashrc_lic
#export LM_LICENSE_FILE=27100@127.0.0.1
```

A .bashrc Example (2)

A screenshot of a terminal window with a blue title bar and standard window controls. The title bar text is 'jiangjunmin@eex001:~'. The terminal content shows the first few lines of a .bashrc file: a shebang line, a comment, and an export statement for the PATH variable.

```
jiangjunmin@eex001:~  
#!/bin/bash  
# Common Path  
export PATH=$APP_DIR/matlab/R2011b/bin:$PATH  
~
```

Commands for Useful Info

* who, who am i, whoami

piano% who

opc	console	Apr 16 08:02	(: 0)
liangzs	pts/2	Apr 16 15:49	(10.14.111.198)
yu jun	dtremote	Apr 18 19:37	(10.14.111.244:0)
wangsj	pts/10	Apr 18 12:47	(10.14.111.132)

* ps

piano% ps

PID	TTY	TIME	CMD
4059	pts/38	0:00	csch
4061	pts/38	0:01	ps
4286	pts/38	0:00	xterm

Manual (1)

* `man`

- * Manual sections

- * `man man`

NAME

`man` - find and display reference manual pages

SYNOPSIS

`man [ - ] [ -adFlrt ] [ -M path ] [ -T macro-package ] [ -s section ] name ...`

`man [ -M path ] -k keyword ...`

`man [ -M path ] -f file ...`

DESCRIPTION

The `man` command displays information from the reference manuals. It displays complete manual pages that you select

Online Manual (2)

by name, or one-line summaries selected either by keyword (-k), or by the name of an associated file (-f). If no manual page is located, man prints an error message.

Source Format

Reference Manual pages are marked up with either `nroff(1)` or `sgml(5)` (Standard Generalized Markup Language) tags.

The `man` command recognizes the type of markup and processes the file accordingly. The various source files are kept in separate directories depending on the type of markup.

Location of Manual Pages

The online Reference Manual page directories are convention-

.....

Online Manual (3)

OPTIONS The following options are supported:

- a Show all manual pages matching `name` within the MANPATH search path. Manual pages are displayed in the order found.
- d Debug. Displays what a section-specifier evaluates to, method used for searching, and paths searched by `man`.
- f `file...`
man attempts to locate manual pages related to any of the given `file`s. It strips the leading path name components from each `file`, and then prints one-line summaries containing the resulting basename or names. This option also uses the win-index database.....

Other Paragraphs in 'man man'

- * USAGE

- * ENVIRONMENT VARIABLES

- * EXIT STATUS

- * SEE ALSO

apropos(1), cat(1), col(1), eqn(1), more(1), nroff(1),
refer(1), tbl(1), troff(1), vgrind(1), whatis(1),
catman(1M), attributes(5), environ(5), eqnchar(5), man(5),
sgml(5)

- * BUGS

Browse through the man Pages

- * **Spacebar** - display next page
- * **b** - previous page
- * **q** - quit

- * **xman** - man has X GUI on Solaris
- * KDE/Gnome help browsers on Linux
- * Web resources

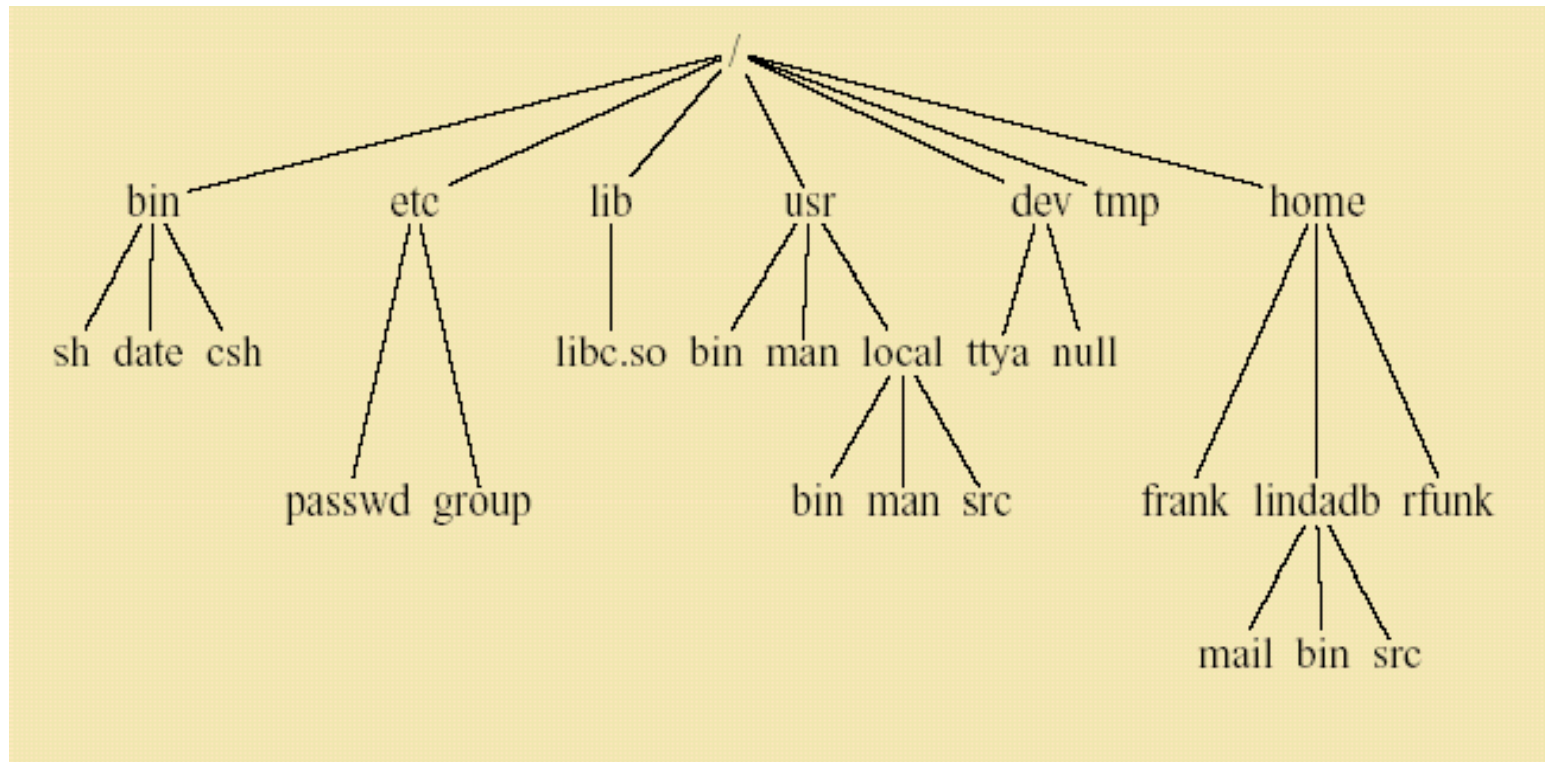
File System



Unix File System

- * Storing files
 - * Text, programs, images, other formatted documents, etc.
- * Being made up of many files
 - * User, Unix System, Unix Commands
- * Generally, *Everything is a file*

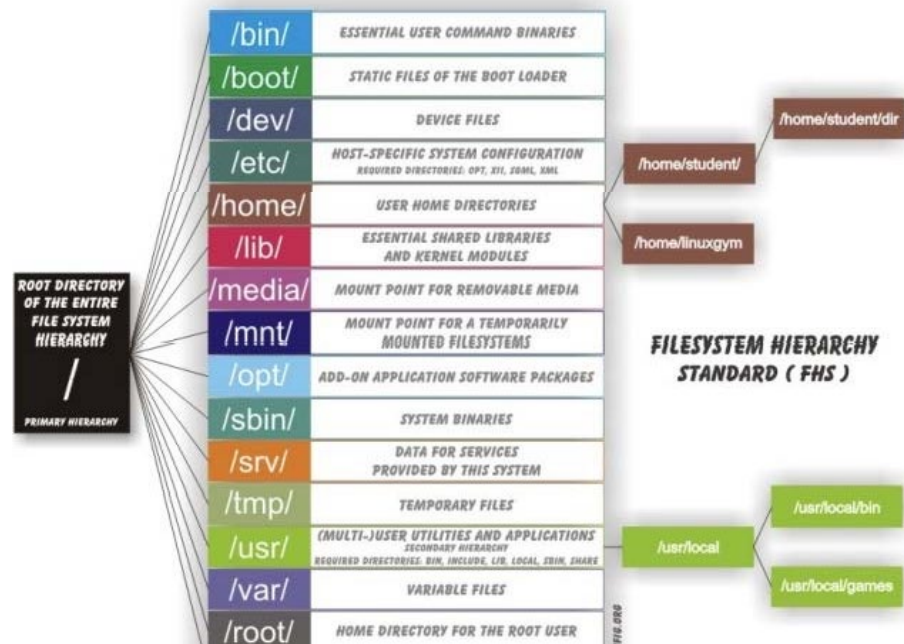
Standard Unix Directories



Some Other Possible Standard

Directories

- * /boot
- * /lost+found
- * /misc
- * /mnt
- * /net
- * /proc
- * /opt
- * /sbin
- * /var



Identifying Files in the Tree

- * Identifying files using full path names
 - * Full path names always begin with /
 - * \$HOME, ~, and EVs
 - * Example: /usr/dt/bin/dtterm
- * Identifying files using relative path names
 - * Dependent on your location in the file system
 - * . and ..
 - * Example: ../../include/define.h

File Structures on Disk

- * Disks mounted to positions in file tree
- * A disk is divided into blocks
 - * 512, 1024, 2048 bytes are common block sizes
- * A disk partition divided into three major sections
 - * superblock
 - * inodes
 - * data blocks
- * More left for later on System Administration

File & Directory Naming Guidelines

- * Don't Use Meta Characters

- * / \ " ` * ; ? { } () [] ~ ! \$ < > | & #

- * <http://en.wikipedia.org/wiki/Punctuation>

- * Do Use

- * a-z A-Z

- * 0-9

- * . _ -

- * Remember Unix is **Case Sensitive!**

Directory Commands (1)

* pwd

```
[jiangjunmin@eex001 ~]$ pwd  
/home/jiangjunmin
```

* cd [directory]

* cd Full Path Names

* cd Relative Path Names

* cd = cd ~ = cd \$HOME

* cd ~thompson

Directory Commands (2)

- * **mkdir** **directory-list**

- * Example: mkdir personal manage

- * **rmdir** **directory-list**

- * Directory must be empty

- * Example: rmdir personal manage

ls – List Contents of Directory

- * Syntax:

- * `ls [ -aAbcCdfFgilMnopqrRstuxl ] [ file ... ]`

- * Frequently used options:

- * `-a` all, including files begin with '.'

- * `-d` directory

- * `-l` long

- * `-F` format

- * `-R` recursive

- * `-t` time

List Directory Contents

- * Try to run 'ls -l'

```
[jiangjunmin@eex001 ~]$ ls -l
total 2292
-rw-r--r--. 1 jiangjunmin jiangjunmin 1262 Jul  6 13:59 bashrc.usr
-rw-rw-r--. 1 jiangjunmin jiangjunmin 75739 Aug  2 13:24 CDS.log
-rw-rw-r--. 1 jiangjunmin jiangjunmin 2246498 Jul 16 13:04 CDS.log.1
drwxrwxr-x. 7 jiangjunmin jiangjunmin 74 May 11 19:02 cqserver
drwxr-xr-x. 2 jiangjunmin jiangjunmin 6 Apr 30 16:31 Desktop
drwxr-xr-x. 10 jiangjunmin jiangjunmin 210 Sep  8 19:19 Downloads
drwxr-xr-x. 27 root root 4096 May 11 14:48 jiangjm
-rw-r--r--. 1 jiangjunmin jiangjunmin 467 Jul  9 01:11 Makefile
drwxrwxr-x. 6 jiangjunmin jiangjunmin 92 Jul 21 21:47 project
drwxrwxr-x. 21 jiangjunmin jiangjunmin 4096 Aug  1 22:41 simulation
-rwxr-xr-x. 1 jiangjunmin jiangjunmin 40 May 11 19:17 vncrun
```

- * Type field (first character)
 - * **d** the entry is a directory;
 - * **l** the entry is a symbolic link;
 - * **b** the entry is a block special file;
 - * **c** the entry is a character special file;
 - * **-** the entry is an ordinary file;

Permissions

- * Access permissions (characters 2-10):
 - * first 3: user/owner
 - * second 3: assigned Unix group
 - * last 3: others
- * Permissions are designated:
 - * **r** read permission
 - * **w** write permission
 - * **x** execute permission
 - * **-** no permission

Other Fields

- * Number of links
- * Owner
- * Group
- * Size or device number
- * Modified time
- * File name

```
drwx----- 54 opc      85      3072 Apr 18 14:4
drwxr-xr-x  11 root    other    512 Aug 15 2001 ../
-rw-r--r--  1 opc     staff     70 Sep 20 2001 ML.log
drwx----- 4  opc     staff    512 Dec  6 22:14 Mail /
drwxr-xr-x   4  opc      85    2048 Apr  1 21:19 Spice/
```

Types Displayed on Linux

* alias ls

```
ls --color=auto
```

blue directories

white/black text files

cyan links

green executables

red compressed archives

pink images

yellow devices

(flashing) red broken links

* ls -F

nothing regular file

* executable file

= socket

directory

@ link

| named pipe

File Maintenance Commands

- * **chmod** change the file or directory access permissions (mode)
- * **umask** get or set the file mode creation mask
- * **chgrp** change the group of the file
- * **chown** change the owner of a file

Change Permissions on File

- * `chmod` [options] file

using + and - with a single letter:

u user owning file

g those in assigned group

o others

- * Examples

`chmod u+w file` gives the user (owner) write permission

`chmod g+rw file` gives the group read/write permission

`chmod go-x file` removes execute permission for group and others

Permission in Numeric

- * `chmod` [options] file

using numeric representations for permissions:

r = 4

w = 2

x = 1

Total: 7

- * `chmod` 7 7 7 filename
 user group others

- * Gives user, group and others the r, w, x permissions by this representation

Permission in Numeric

- * `chmod 750 filename`

- * gives the user read, write, execute
- * gives group members read, execute
- * gives others no permissions

- * `chmod 640 filename`

- * gives the user read, write
- * gives group members read
- * gives others no permissions

Setting Default Permissions

- * `umask mask`

- * set in startup files for the account
- * masks out permissions
- * umask numbers added to *desired permission number* equals 7

- * Example

`umask 022`

means default permission of file is 755, that is, `rwxr-xr-x`

Display Commands

- * **echo** echo the text string to stdout
- * **cat** concatenate (list)
- * **head** display first 10 (or #) lines of file
- * **tail** display last 10 (or #) lines of file
- * **more** browse or page through a text file
 - * **SPACE RETURN b q**

File Manipulation Commands

- * **cp** copy file
 - * **cp** [-fipRr] source_file target_file
- * **mv** move (or rename) file
 - * **mv** [-fi] source target_file
 - * **mv** [-fi] source ... target_dir
- * **rm** remove (delete) a file (directory)
 - * **rm** [-f] [-i] file ...
 - * **rm** -rR [-f] [-i] dirname ... [file ...]
- * Meaning of each option here

ln and unlink

- * **ln** [options] source target - Link to another file
 - * symbolic link
 - * hard link
- * **unlink** filename - Remove the link

find - Find Files

* **find** **directory-list** **[options]** **[actions]**
[...]

- * `find . -name Net\*.c -print`
- * `find . -newer empty -print`
- * `find /usr/local -type d -print`
- * `find . \(user Bill -o -size +10000c\) -print -
exec rm { } \;`
- * `find ~Bill ~Denis -size +1000 -atime 30 -ok rm
{ } \;`

Compression/decompression

- * `compress` [-cfv] [-b bits] [file]

- * `uncompress` [-cfv] [file ...]

- * .Z as suffix

- * `gzip` [-cdfhlnNrtvV19] [-S suffix] [file ...]

- * `gzip filename` filename->filename.gz

- * `gzip -d filename` filename.gz->filename

- * `gzip -h` help for gzip

- * `gunzip`

- * `bzip2 / bunzip2`

Utility - wc

- * **wc** - display a count of lines, words and characters in a file

- * -c

- * -l

- * -W

- * `ls | wc -l`

Utility - diff

- * `diff file1 file2`

- * Line-by-line differences between pairs of text files
- * Often used for comparing text file (source code, netlist, data list, etc.) in different versions.
- * `tkdiff` vs. `windiff`

Sample Output of tkdiff

```
[meld] meld : [dev] meld - Meld
File Edit Settings Help

meld : src [meld] meld : [dev] meld

/home/stephen/meld/meld /home/stephen/dev/meld/src/meld

31# 19#
32sys.path += [ #LIBDIR# 20# i18n support
33] 21#
34import paths 22import program
35import gettext 23import paths
36import locale 24import gettext
37 25import locale
38try: 26
39    locale.setlocale(locale.LC_ALL, '') 27try:
40    gettext.bindtextdomain("meld", paths.loc 28    locale.setlocale(locale.LC_ALL, '')
41    gettext.textdomain("meld") 29    gettext.bindtextdomain(program.name, path
42    gettext.install("meld", paths.locale_dir 30    gettext.textdomain(program.name)
43except (IOError,locale.Error), e: 31    gettext.install(program.name, paths.local
44    # fake gettext until translations in pla 32except (IOError,locale.Error), e:
45    print "(meld): WARNING **: %s" % e 33    print "(%s): WARNING **: %s" % (program.n
46    __builtins__.__dict__["_"] = lambda x : 34    __builtins__.__dict__["_"] = lambda x : x
47    __builtins__.__dict__["ngettext"] = gettext.i 35
48 36#
49# 37# python version
50# python version 38#
51# 39import sys
52 40pyver = (2,3)
53pyver = (2,3) 41pygtkver = (2,4,0)
54pygtkver = (2,6,0) 42
55 43def ver2str(ver):
56def ver2str(ver): 44    return ".".join(map(str,ver))
57    return ".".join(map(str,ver)) 45
58 46if sys.version_info[:2] < pyver:
47    print "!!%s requires %s/%s" % (program.name, pyver, pygtkver)
```

Utility - grep

- * **grep/egrep/fgrep** - search the argument for all occurrences of the search string (get REP)
 - * The **grep** utility is used to search for regular expressions in Unix files.
 - * **fgrep** searches for exact strings (not pattern).
 - * **egrep** uses "extended" regular expressions.
- * **grep regexp file**
 - * `grep static ./*.c`
 - * `grep ^# *.c`
 - * `man ls | grep -ci permission`

grep a Word in a File

```
piano% grep -ni Event main.c
```

=> Check for output from real example

Question

? What's the difference of `grep` and `find`?

--- An interconnecting network of the commands and utilities

Utilities

- * **file** - determine file type
- * **sort** - sort, merge, or sequence check text files
- * **which** - locate a command; display its pathname or alias
- * **whatis** - search the whatis database for word
- * **apropos** - search the whatis database for string
- * **spell** - report spelling errors of file
- * **look** - find words in dictionary
- * Other standard utilities in /usr/bin, /usr/sbin
- * See http://en.wikipedia.org/wiki/List_of_Unix_utilities